BRIELLE JOHNSTON

AI and Data Science Developer

briellejohnston2@gmail.com · (438) 936-4050 ·

SUMMARY

AI and Data Science Developer with over six years of experience across predictive modelling and generative AI. Leads the design and delivery of production systems, from model development through to deployed applications, for clients in consulting, cybersecurity, and government.

LANGUAGES

LANGUAGES: English (Native), French (Native)

EDUCATION - EASTERN MICHIGAN UNIVERSITY

Eastern Michigan University — B.Sc. Actuarial Science **2016 – 2020**

- Academic Scholarship
- Athletic Scholarship on the Swim&Dive team
- 3.86/4.0 GPA

TECHNICAL SKILLS - AI & GENERATIVE AI

AI & Generative AI: Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Context Protocol (MCP), Semantic Kernel, AutoGen, LangChain, LlamaIndex, and related AI orchestration technologies.

TECHNICAL SKILLS - MACHINE LEARNING & DATA SCIENCE

Machine Learning & Data Science: Scikit-learn, PyTorch, Classification, Regression, Time-Series Forecasting, Feature Engineering, Hyperparameter Optimization, Model Evaluation, and related machine learning technologies.

TECHNICAL SKILLS - PROGRAMMING LANGUAGES & FRAMEWORKS

Programming Languages & Frameworks: Python, SQL, TypeScript, FastAPI, Flask, React.

TECHNICAL SKILLS - CLOUD & DATA PLATFORMS

Cloud & Data Platforms: Azure AI Foundry, Azure OpenAI, Azure AI Search, Azure Machine Learning, Cosmos DB, Azure SQL Database, Azure Functions, App Services, Databricks

TECHNICAL SKILLS - DEVOPS & SOFTWARE ENGINEERING

DevOps & Software Engineering: Docker, Git, CI/CD, REST APIs, Cloud-Native Frontend and Backend Application Development.

CERTIFICATIONS

Certifications: Databricks Data Engineer Associate; Databricks Machine Learning Associate.

WORK EXPERIENCE - BDO

BDO **2023 – Present**

SENIOR AI AND DATA SCIENCE DEVELOPER

- Designed and deployed agentic AI solutions using Semantic Kernel, Azure AI Foundry, MCP, n8n, and related agentic AI technologies, enabling multi-agent collaboration, enterprise system integration, dynamic workflow orchestration, and autonomous task execution.
- Designed and implemented production Retrieval-Augmented Generation (RAG) systems using Azure OpenAI, Azure AI Search, FastAPI, and Docker, leveraging prompt optimization, retrieval tuning, and automated evaluation workflows to deliver citation-grounded enterprise knowledge retrieval solutions.

- Developed intelligent document-processing solutions using document intelligence, OCR technologies, LLMs, and LlamaIndex to extract, validate, and transform unstructured content into structured business data and automate document generation.
- Built and deployed production AI applications using Python, FastAPI, Flask, React, Docker, Azure Container Apps, and App Services, implementing cloud-native architectures, CI/CD pipelines, and automated deployment strategies.
- Developed and optimized machine learning and deep learning solutions for classification, regression, and time-series forecasting to predict factory incidents, machinery overloads, and sensor values. Applied advanced feature engineering, statistical preprocessing, hyperparameter tuning, custom loss functions, and model evaluation techniques to improve model performance and reliability.
- Provided technical leadership through architecture reviews, pull request reviews, technology evaluations, and collaborative solution design, helping teams scale AI and data-driven solutions from proof-of-concept to production.
- Conducted technical interviews, mentored junior team members, facilitated knowledge-sharing sessions, and collaborated with stakeholders to identify business requirements and recommend appropriate AI and technology solutions.

WORK EXPERIENCE - QOHASH INC.

Qohash Inc. **2022 – 2023**

DATA SCIENTIST

- Built several machine learning models from inception to completion, encompassing tasks such as data preparation, augmentation, preprocessing, tokenization, model training, hyperparameter tuning, and testing.
- Created a text classifier that not only gave the prediction, but also the words in the input text that had the largest impact on the prediction, allowing for a more understandable model for the clients.
- Constructed an Optical Character Recognition (OCR) system to read text from images and documents containing embedded images. Added a model to the OCR pipeline that would classify whether an image contained text or not.
- Collaborated with software developers to efficiently implement machine learning models to the product.

WORK EXPERIENCE - PRIVY COUNCIL OFFICE

Privy Council Office **2020 – 2022**

DATA SCIENTIST

- Utilized Python and R to preprocess vast datasets and create compelling visualizations to aid executives in decision-making.
- Employed text mining techniques to extract valuable insights from survey text responses and conduct web scraping to seamlessly acquire data from various online sources.
- Created databases, modified them using SQL and used them to increase efficiency.
- Prepared, combined and cleaned large amounts of data in Tableau Prep and Power Queries.
- Created powerful dashboards in Tableau and Power BI.